

Assignment11 – Natural Language

Given: July 1 Due: July 6

Problem 11.1 (Language Models)

Consider a language L over the alphabet with characters a, b, c, d , and consider a corpus D for it consisting of the five words

$abccc \quad abbd a \quad a a b c \quad b c d b \quad a a b a c$

You want to build a 3-gram model for L .

1. How many 3-grams are there over L ?
2. Now assume you store the 3-gram model as a Bayesian network with variables w_1, w_2 , and w_3 (representing the 3-gram $w_1 w_2 w_3$), how many entries does the conditional probability table for w_3 have?
3. Give the numerical value of $P(w_3 = a \mid w_{1:2} = aa)$ in the resulting model.
4. Give the numerical value of $\text{tf}(a, cdad)$.
5. Give the numerical value of $\text{idf}(a, D)$.
6. Now you want to use the tfidf vector embedding. What are the objects that are embedded, and how are the vectors computed using tf and idf ?
7. Now assume you are interested in all documents that contain the subword bc . A query system returns all 5 documents.
Give the numerical values of

1. precision:

2. recall:

Problem 11.2 (Information Retrieval)

1. Explain (in about 2 sentences) the connection between tfidf and *cosine similarity* to relate texts.
2. Explain (in about 2 sentences) the difference between *information retrieval* and *information extraction*.
3. Explain (in about 3 sentences) what kind of function is learned in the continuous *bag of words* algorithm, as used e.g. in *Word2Vec*, and how the training proceeds.

Problem 11.3 (Grammars)

Consider the following *probabilistic grammar*:

<i>S</i>	→	<i>NP VP</i> [1]
<i>NP</i>	→	<i>Article Noun</i> [0.6] <i>Name</i> [0.4]
<i>VP</i>	→	<i>Verb</i> [0.5] <i>TransVerb NP</i> [0.5]
<i>Article</i>	→	the[0.7] a[0.3]
<i>Noun</i>	→	stench[0.2] breeze[0.3] wumpus[0.5]
<i>Name</i>	→	John[0.3] Mary[0.7]
<i>Verb</i>	→	smells[1]
<i>TransVerb</i>	→	sees[0.6] shoots[0.4]

1. Using this *grammar* as an example, explain the difference between *structural rule* and *lexicon*.

2. Give the *probability* of the *sentence*

John sees the wumpus

(You have to give the *expression* with concrete *values* plugged in, but you do not have to *compute* the result.)

3. Now assume we do not know the *probabilities* of the *production*, and our *corpus* is

John sees the wumpus. The wumpus smells. John shoots the wumpus.

Give the *probability* that we can learn for $NP \rightarrow Name$ from this *corpus*.